Report

Project Title: Exploratory Data Analysis of AI & Social Media Impact on Student Health and Academic Performance

Target Platform & Dataset: Python (Pandas, Seaborn, Matplotlib) & Power BI Desktop

Prepared for: Academic Project Documentation & Presentation

1. Project Overview & Objectives

This project presents a structured Exploratory Data Analysis (EDA) investigating the relationships between daily social media usage, AI tool adoption, physical activity, sleep patterns, and overall physical and mental health scores among students.

Key Objectives:

- **Identify Risk Factors:** Evaluate how digital habits (social media and AI tool usage) impact student well-being (mental and physical health scores).
- **Examine Protective Behaviors:** Quantify the extent to which physical activity and sleep mitigate negative health outcomes.
- **Assess Demographic Differences:** Determine whether digital adoption patterns or health metrics vary significantly across academic levels (High School, College, University), age groups, or genders.
- **Transition to Interactive BI:** Establish a step-by-step workflow to transition data and embed visualizations into a Power BI dashboard for real-time reporting.

2. Methodology & Dataset Summary

- **Dataset Scale:** 16,000 student records spanning diverse demographics.
- **Core Variables Analyzed:**
    - Age (Numerical: 14 to 24 years)
    - Gender (Categorical: Male, Non-binary, Female)
    - Education_Level (Categorical: High School, College, University)
    - Daily_Social_Media_Hours (Numerical: Daily screen time duration)
    - Daily_AI_Tool_Usage_Hours (Numerical: Daily AI assistant usage)
    - Sleep_Hours (Numerical: Nightly rest duration)
    - Physical_Activity_Hours (Numerical: Daily exercise duration)
    - Mental_Health_Score (Numerical: Standardized 0–100 well-being scale)
    - Physical_Health_Score (Numerical: Standardized 0–100 health scale)

3. Detailed Visualizations & Analytical Interpretations

Chart 1: Physical Activity vs. Physical Health Score (Scatter Plot by Education Level)

- **What it compares:** Physical activity hours (X-axis) against physical health scores (Y-axis), colored by education level.
- **Analytical Findings:**

    - **Positive Trajectory:** As physical activity increases from 0 to 5 hours, the minimum physical health baseline moves upward from approximately 48 to nearly 90 points, proving that higher exercise levels support better physical health.
    - **Education Uniformity:** Distributions for High School, College, and University students mix evenly across all activity levels, demonstrating that academic tiers share identical health-to-exercise patterns.

Chart 2: Correlation Heatmap Across Numerical Features

- **What it shows:** A grid matrix displaying linear correlation coefficients from -1.00 to +1.00 across all numerical student variables.
- **Analytical Findings:**
    - **Social Media Impact:** Strong negative correlations exist between daily social media duration and Mental_Health_Score (**-0.40**) as well as Physical_Health_Score (**-0.31**).
    - **Exercise & Health:** Physical_Activity_Hours shows a positive correlation with Physical_Health_Score (**+0.41**).
    - **Sleep Interdependence:** Sleep_Hours positively correlates with Mental_Health_Score (**+0.30**), while excessive social media negatively impacts sleep duration (**-0.29**), creating a multi-variable chain reaction.
    - **Age Independence:** Age shows negligible linear correlation with digital habits or health scores (**0.00** to **0.01**).

Chart 3: Screen Time Habits Across Academic Levels (Violin Plot)

- **What it compares:** Daily social media hours across different education levels using a density violin plot combined with an inner box summary (inner="box").
- **Technical Parameter Explanation (inner="box"):** It embeds a miniature box plot inside the violin curve to cleanly display the median (white marker) and the interquartile range (thick central bar representing the middle 50% of students).
- **Analytical Findings:**
    - **Consistent Medians:** All three academic tiers exhibit an identical median daily screen time of approximately 4 to 5 hours.
    - **Uniform Spread:** The central bulk of students across High School, College, and University cluster tightly between 3 and 6 hours of daily social media usage.

Chart 4: AI Tool Adoption Differences by Gender (Strip Plot)

- **What it compares:** Daily AI tool usage hours across gender identities (Male, Non-binary, Female) using jittered individual data points.
- **Analytical Findings:**
    - **Equitable Usage:** Horizontal bands of individual data points stack evenly from 0 to 10 hours across all gender groups, showing no gender bias in AI technology adoption.
    - **Integer Reporting:** Clustered points at whole-number intervals indicate students typically log their AI tool usage in full-hour increments.

Chart 5: Age Distribution of the Student Sample (Histogram with KDE)

- **What it shows:** The demographic spread of student ages accompanied by a smooth Kernel Density Estimation (KDE) curve.
- **KDE Definition:** Kernel Density Estimation takes blocky histogram bars and converts them into a smooth curve to visualize data density, probability concentration, and peak trends.
- **Analytical Findings:**
    - **Core Sample Uniformity:** Students aged 15 to 23 exhibit a uniform distribution, maintaining a steady count of approximately 1,100 to 1,200 records per age tier.
    - **Boundary Spikes:** Pronounced spikes occur at the boundary ages of 14 and 24, reaching nearly 2,500 records each.

Chart 6: Sleep Duration vs. Mental Health Score (Line Plot)

- **What it tracks:** The trend of average mental health scores as nightly sleep hours fluctuate.
- **Analytical Findings:**
    - **Positive Trend Line:** An upward slope demonstrates that higher nightly sleep hours correspond to improved mental health scores, while lower sleep hours highlight vulnerabilities to mental fatigue.

Chart 7: Screen Time vs. AI Usage Faceted by Gender (Facet Grid)

- **What it compares:** Multivariate relationships between social media hours and AI tool usage, split into side-by-side panels by gender.
- **Analytical Findings:**
    - **Multi-Variable Independence:** Side-by-side scatter grids confirm that high social media consumption and high AI tool adoption exist independently across all gender subsets without major skewing.

4. Power BI Dashboard Integration Workflow

To integrate the analysis into a Power BI professional reporting environment:

1. **Export Cleaned Data:** Save the processed pandas DataFrame into a structured CSV file format (.csv) for seamless database reading.
2. **Import into Power BI Desktop:** Open Power BI Desktop, select **Get Data**, load the CSV file, and verify the columns in the fields pane.
3. **Recreate Interactive Visuals:** Utilize Power BI native visuals (Matrix for correlations, Clustered Column charts for age distribution, Box and Whisker components for gender/AI metrics) to match Python outputs.
4. **Implement Slicers:** Add slicers for Education_Level and Gender to enable dynamic real-time filtering across all dashboard metrics simultaneously.
5. **Publish & Share:** Save the .pbix report file and publish it to the cloud service for collaborative review.

5. Strategic Recommendations & Project Conclusion

- **Digital Wellness Programs:** Given the negative correlation between social media hours and mental health scores (**-0.40**), institutions should introduce screen-time awareness initiatives.
- **Active Lifestyle Promotion:** Encouraging daily physical activity successfully mitigates physical health declines, as proven by the positive correlation (**+0.41**).
- **Sleep Hygiene Focus:** Prioritizing healthy nightly rest counters screen-time-induced sleep deficits, directly supporting student mental well-being.